Readout Circuit and Signal Generator for Electron Beam Characterisation

IMG-Current Report

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Abstract

In electron microscopy, efforts are underway to improve image contrast through the implementation of an electron gun. The electron guns purpose is to generate an electron beam characterized by two specific parameters, beam current and beam width, to create an electron phase plate which enhances contrast. The current state of developments require a method of measuring the electron beams current and width in order to control its performance. The thought out method is to periodically pass the electron beam though a rotating disk into a Faraday cup, connected to a circuit that measures the current and beam width from the signals shape. The shape can be quantified by the signals rise time and the derivatives FWHM. The task for this paper was to design a circuit that can perform this measurement and to test its accuracy for 10 μA, 5 μA and 2.5 μA. The results show that the circuit's measurements were off by 18%, 5% and 32% for the rise time and 52%, 29% and 6% for the FWHM for 10, 5 and 2.5 μA respectively. The errors in the circuit measurements are believed to arise from signal artifacts which become prominent at different current magnitudes due to component performance. Errors are also believed to be related to the oscilloscopes sampling rate of only 100 MHz. Increasing it could resolve issues related to sampling errors and signal filtering problems. It is also believed that some errors occur due to impedance mismatching in the circuits transmission line. To conclude, the circuit fulfills the aim and is able to measure an electron beam with currents between 10-2.5 μA, but more improvements are necessary for better and more reliable results and performance.

Keywords

Electron Beam, Signal Generator, Faraday Cup, Filter Design, Current Detection.

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Acronyms

ADC = Analog to digital converter
API = Application Programming Interface
AWG = Arbitrary wave generator
CSV = Comma Separated Values
FIR = Finite Impulse Response
FWHM = Full width at half maximum
OP-Amp = Operational amplifier
RBA = Responsible Business Alliance
SDK = Software Development Kit
SNR = Signal to Noise Ratio

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Chapter 1

Introduction

Electron microscopy is a powerful tool that is used for detecting pathogens and understanding the microscopic construction of the cell [1]. This can further be used to better understand and treat potentially lethal bacterial and viral diseases. One of the important parameters that quantify the quality of an electron microscopic image is the image contrast. The contrast of an image is the ability to differentiate aspects of the object from its background. This contrast can be achieved by several different methods, one of which is by using a phase plate.

A problem with the traditional phase plate is the effect of charging which leads to additional noise in the image [2]. Koeck [3] suggests a new type of phase plate that consists of an electron beam with a specific charge density (current and radial distribution) which accounts for this charge buildup.

The electron gun, used to create this electron beam phase plate, is currently under construction and to make sure that it produces a beam with the correct charge density according to Koeck's calculations [3], the current and radius of this beam must be measured. The current of the beam should ideally be in the range of 1-10 μA and the diameter should be around 1 μm [3].

To measure the characteristics of this beam, a rotating disk with a cutout and an angular velocity of 1 m/s can be put in the focal point of the beam and a Faraday cup can be placed roughly 15 cm behind the focal point to measure the current from the beam. A Faraday cup is a device that can capture electrons and release a current directly proportional to the current of the electron beam [4]. The idea is that the rotating disk with the cutout periodically reveals and blocks the beam causing the current measured

at the Faraday cup to periodically rise and drop. By analyzing the rise of the current, which should be directly related to the radial distribution of the electrons in the beam, one should be able to accurately characterize the electron beam in its focal point. An illustration of this setup can be seen in figure 1.0.1.

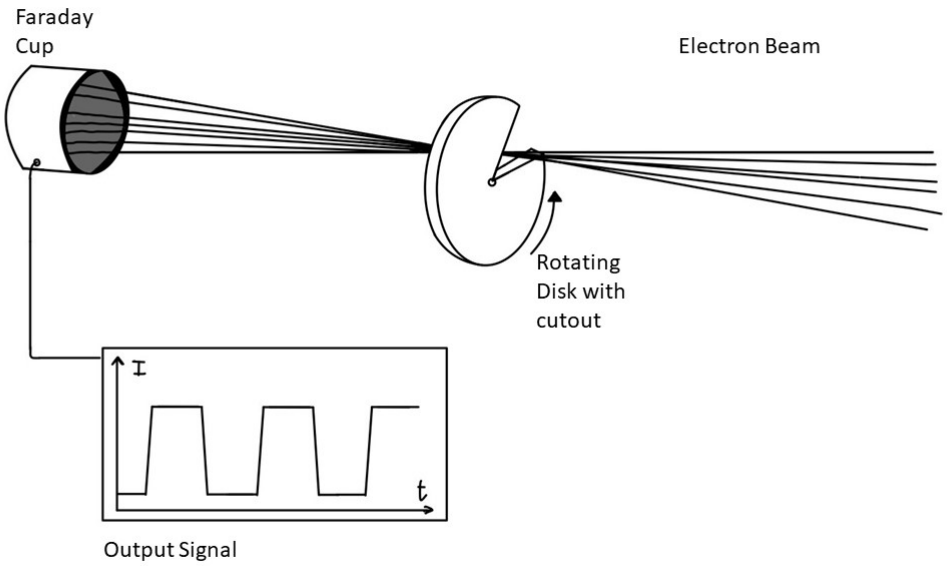


Figure 1.0.1: Illustrative image of faraday cup experimental setup

Since the Faraday cup is not yet constructed, a signal generator can be created that mimics the Faraday cup's output signal. This signal would be a step response with a specific slope. The slope can be characterised by looking at it's structure and in particular it's rise time. If a circuit can measure the signal produced by this generator, and it's structure can be analyzed, then the same circuit should be able to measure the output from the Faraday cup in the real experimental setup.

1.1 Task

The task for this project was to design a circuit that gives the current readout from the Faraday cup. Instead of using the real setup with an electron beam and the Faraday cup, a signal generator was created to generate the incoming signal from the Faraday cup into the circuit.

1.2 Aim

The aim of this project was to design and test a signal generator that mimics the signal from a Faraday cup and to design and test a readout circuit that can accurately measure the generated signal. The goal for this project was to

- create a measuring circuit and a signal generator that produces a square-wave signal with a 1 μs rise time
- test the measuring circuits accuracy for 10 μA, 5 μA and 2.5 μA

Chapter 2

Theoretical Background

This chapter provides background information about electrical components, electronic difficulties and electrical parameters to give a better insight into the methodology and core problems of the project. This chapter is split into four main parts with subsections going further into these areas.

2.1 Measure Beam Current and Width

Measuring the beam current can be done by looking at the magnitude of the outputted voltage signal from a Faraday cup over a resistor and using Ohm's law to find the current. The width of the beam can then be estimated by looking at the rise time of this signal, and knowing the speed of the rotating disk we can multiply the rise time and the rotational velocity of the disk to get the estimated diameter of the beam.

2.1.1 Faraday cup

A Faraday cup is a device that can measure a particle beam by capturing the charged particles and then discharging a signal proportional to the amount of particles captured. A Faraday cup normally consists of a cylindrical lead tube with a surrounding aluminium casing and a stainless-steel core inside which absorbs most of the incident particles, see Figure 2.1.1 [5].

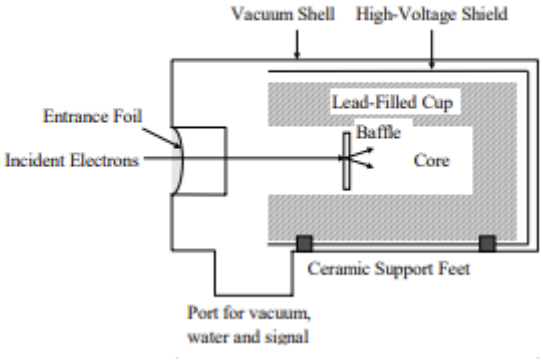


Figure 2.1.1: Schematic of a Faraday cup [5]

As charged particles enter the aperture of the cup they remain inside the cup resulting in a buildup of electrical charge [6]. The accumulated charge can then be discharged through either a resistor or a circuit which creates an electrical current that is proportional to the amount of charges captured.

2.1.2 Experimental Setup

To measure the width of an electron beam, it can be injected into a Faraday cup through a rotating disk with a circle sector cutout. The electron flow will be completely blocked by the disk, except when it passes through the cutout. It will then be injected into the Faraday cup to generate a signal.

When the electron flow gradually reaches the edge of the cutout, some electrons will begin to inject into the Faraday cup, and the detected current will start to gradually increase. When the current reaches the maximum, the electron beam will be completely in the cutout.

The relationship between the distribution along the radial direction of the beam and the current can therefore be given by the following equation:

$$\begin{aligned} \frac{dI}{dr} &\propto N(r) \\ dr &= vdt \end{aligned} \tag{2.1}$$

Where dt is an infinitesimal change in time, v is the rotating speed of the disk, $\frac{dI}{dr}$ is the change in current detected with the change in radial distance revealed by the disk and

$N(r)$ is the radial distribution of the electrons.

By finding $N(r)$ one can quantify the radial distribution of the particles. This can then give a good estimate of the radius of the beam by calculating the FWHM (Full width at half maximum) from the distribution [7].

2.2 Noise and Filters

This section discusses the importance of understanding how noise is usually generated in circuits. It also provides insight on how this noise can be quantified and dampened through the use of both analog and digital filters. It finally talks about impedance mismatching, which is a common cause for signal distortion in high impedance circuits.

2.2.1 Component Noise

When designing a circuit and choosing relevant components it shall be taken into account that every addition of a component to the circuit will also be an addition of noise. To what extent and how this noise will behave is often component-dependent, although there are some universal effects. Depending on the contact between the components in a circuit the resistance can vary. This phenomenon is usually referred to as contact noise. To avoid this it is important to perform good soldering connections [8].

The electric flow is generated by discrete electrons, with electric charges. Thermal noise is a result of random movement by discrete electrons due to thermal energy and results in voltage fluctuations [9]. It is therefore of interest to evaluate the generated heat when deciding on a resistor since a high resistance increases the heat [10].

A similar noise that affects low-current circuits is shot noise. This is a result of the discrete flow of electrons and is therefore more noticeable with the presence of fewer electrons, in weaker currents. The electric signal can then contain a spiky irregular pattern [11].

Another interference that can occur is when capacitors are exposed to mechanical vibration that changes the electric field of the component. This behavior is known as piezoelectric noise and is more common for ceramic capacitors where the results are

small voltage fluctuations [12].

The presence of inductance and resistance in capacitors and wires can result in voltage spikes during rapid changes. This behavior is referred to as parasitic inductance or resistance and appears as abrupt, irregular voltage fluctuations or spikes in response to rapid changes [13]. In addition, surrounding electromagnetic fields can be detected and interfere with the original signal. It is therefore of value to evaluate the effects of common electromagnetic noise and consider the wavelength of radio signals and power lines [14] when designing a circuit.

2.2.2 Signal-to-Noise-Ratio

As explained in the previous section, electrical signals and electrical components will in general not work ideally and instead signals will be complex and contain a lot of noise. Signal-to-noise-ratio (SNR) is a metric used to compare the strength of the desired signal to the level of background noise. A higher SNR indicates a stronger, more reliable signal, which can be crucial for achieving robust and accurate transmission of information. If the SNR value is above 1 then the signal can be distinguished from the noise, otherwise the noise is larger than the signal. It can be expressed as the ratio of the power or amplitude of a signal to the power or amplitude of the noise present in the signal, see Equation 2.2 [15].

$$SNR = |\frac{A_{signal}}{A_{noise}}|^2 \tag{2.2}$$

Depending on the type of signal the amount of noise can differ but overall it is essential to achieve a good SNR.

2.2.3 Active Filter

In electrical circuits filters can be used to reduce noise and manage the rise time of electrical signals. These parameters are both a result of the chosen resistor and capacitor used to design the filter [16]. The cutoff frequency, f_c, for what frequency the signal begins to attenuate is determined by the equation 2.3.

$$f_c = \frac{1}{2\pi RC} \tag{2.3}$$

Where R is the resistance value in Ω and C is the capacitance in Farad. By using this filter unwanted noise at a certain frequency threshold becomes less noticeable.

The time constant, τ, of the filter is determined by equation 2.4.

$$\tau = RC \tag{2.4}$$

This constant has an impact on a filter's signal response and is defined as the time it takes for the filtered signal to reach 63.2% of its maximum. By strategically choosing the R and C values, a response with a certain rise time can be achieved. If the rise time, τ_r, is defined as the time taken for the signal to go from 10% to 90% of its max, then the following relationship exists between τ and τ_r for a step function input signal [17]:

$$\tau_r = \ln 9\tau \approx 2.2\tau \tag{2.5}$$

An electrical filter can either be passive or active. Active filters, that utilize operational amplifiers (OP-amp), can provide enhanced flexibility and precision in shaping electrical signals [16]. Passive filter's performance is affected by the total impedance of the entire circuit whereas active filters have the benefit of isolating the circuit impedance from the filter's impedance. This means that in theory an active filter's performance is never compromised. Figure 2.2.1 shows a schematic of how a first order active low-pass filter can look like.

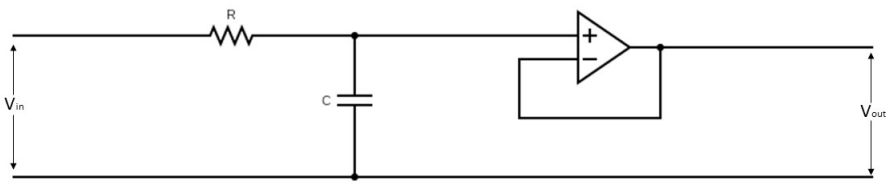


Figure 2.2.1: Schematic of an active low-pass filter

2.2.4 FIR Filters

A finite impulse response (FIR) filter is a kind of filter where the impulse response is of finite duration since it settles to zero in finite time [18]. FIR filters can be both discrete-time or continuous-time and be digital or analog [18]. For a causal discrete-time FIR filter of order N, each value of the output sequence will be a weighted sum of the most recent input values:

$$y[n] = \sum_{i=0}^{N} b_i \cdot x[n-i] \tag{2.6}$$

where $x[n]$ is the input value, $y[n]$ is the output signal, N is the order of the filter and b_i is a coefficient. The operation described in Equation 2.6 is known as a convolution and $x[n-i]$ is commonly referred to as number of taps. The more number of taps, the higher the filters order is. The number of taps must be odd if the pass-band of the filter includes the Nyquist frequency [19].

FIR filters can be designed to be both low-pass and high-pass filters. Since the convolution operation is equivalent to multiplication in the Fourier space [20], one can think of the filter as scaling wanted and unwanted frequencies accordingly. FIR filters can be digitally implemented in python using the scipy python library [19].

2.2.5 Impedance Mismatching and Reflections

When current travels through a circuit, the voltage is generated as an effect of the electromagnetic field that is placed onto the circuit according to the following equation:

$$V_{BA} = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l} \tag{2.7}$$

Where $\vec{E}$ is the electric field, $d\vec{l}$ is the infinitesimal line element across the circuit and V_{BA} is the voltage across some arbitrary points B and A on the circuit.

This implies that the signal sent across the circuit is in-fact an electromagnetic wave with some wavelength λ and some phase ϕ. Due to this effect, if a wave goes from travelling through a low impedance node to a high impedance node a large amount of the wave will be reflected back in the direction it came from [21]. The proportion of

the wave reflected, Γ, is given by the following equation:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{2.8}$$

Where Z_L and Z_0 are the load impedance and the circuit impedance respectively.

This implies that if the load impedance is larger than the circuit impedance a large proportion of the wave is reflected back towards the circuit. This can lead to reflection artifacts which distort the signal generated by the circuit, often as a form of some type of damping or overshoot [21].

To compensate for this, by looking at equation 2.8, one can choose the circuit's impedance in such a way that $Z_L = Z_0$ so that Γ is 0. This would mean that no reflections occur in the circuit and the reflection artifact is entirely removed.

2.3 Oscilloscope

An oscilloscope is a type of electronic instrument that can be used to visualise high-speed changing electronic signals. The signals can be amplified in amplitude and scaled in time to make both low and high frequency signals visible [22]. This means that an oscilloscope is a useful tool that is usually used when measuring electronic signals.

Nowadays, oscilloscopes are generally digital oscilloscopes. This means that they use analog-to-digital converters (ADCs) to convert the analog signal from the input channels into digital numbers [23]. These numbers are proportional to the incoming voltage and are based on the bit-rate of the oscilloscope [23].

To keep the signal stable on the screen for analysis, modern oscilloscopes have triggered sweeps. The triggered sweep starts from the chosen point of the signal and sweeps in from that point in a chosen direction or in both directions simultaneously. This is a way to collect specific data from an incoming signal.

The measurement of a signal using an oscilloscope is often done by using it together with an oscilloscope probe. This probe usually has a inner resistance and capacitance that stops it from heavily distorting the signal [24]. Although this generally helps to correctly collect the signal it can in some situations lead to noise and artifacts in the acquired signal.

2.3.1 PicoScope 2204A

The PicoScope 2204A is a portable oscilloscope developed by Pico Technology. This oscilloscope has 2 channels and one AWG (arbitrary wave generator) [25]. It has a sampling rate of 100 MHz (100 MSamples/s) and can be connected to a computer with USB [25].

The PicoScope 2204A has a desktop application that allows one to view the signal received from the various channels of the oscilloscope but it also has a SDK (Software Development Kit) [25]. The SDK allows a user to interact with the oscilloscope using a programming interface with the users preferred programming language. It is in essence an API (Application Programming Interface) that allows one to program the internal circuit of the oscilloscope. The SDK has many different functions of which the following are the most relevant to this project [26]:

- **ps2000_open_ unit:** This function is used to open communication with the oscilloscope.
- **ps2000_set_channel:** This function chooses the channels on the oscilloscope that are to be used for acquiring data. It is also in this command that the window range is selected, i.e. how big the window should be to correctly collect the entire signal. One also chooses if the oscilloscope is to use AC or DC coupling. AC coupling removes all DC components from a signal whereas DC coupling allows all signal components to pass through.
- **ps2000_set_trigger:** This function sets the trigger of the oscilloscope. It tells the oscilloscope when the oscilloscope should start acquiring data and how much data should be collected, both before and after the trigger event.
- **ps2000_get_timebase:** This function takes the desired sampling rate as one of it's inputs and depending on this and the number of samples wanted it returns a timebase for the oscilloscope to use when collecting data. A timebase effectively sets the sampling rate or temporal resolution of the oscilloscope based on the preset sampling rates that the specific oscilloscope allows
- **ps2000_run_block:** This function takes the above set parameters into account and sends an API call to the oscilloscope to start collecting data.
- **ps2000_ready:** This function tells the program when the oscilloscope has

finished collecting the data from the *ps2000_ run_ block* command.

- **ps2000_get_times_and_values:** This function returns a two dimensional array containing the ADC values and the times at which the values were acquired from the signal.
- **ps2000_stop:** This function tells the oscilloscope to stop collecting data.
- **adc2mV:** This function converts the ADC value to mV (millivolt) values.

2.4 Components

This section talks about the background relating to the relevant components used in this project. It provides insight as to why one would choose to incorporate these components into a circuit and how different parameters of the components affect the circuit's performance.

2.4.1 Operational Amplifier

The operational amplifier is an electric component that can amplify an electrical signal. The amplification can be described by the following equation.

$$\text{Gain} = \frac{V_{out}}{V_{in}} \tag{2.9}$$

Where V_{in} is the voltage across the input nodes of the amplifier and V_{out} is the output voltage from the output of the amplifier to the ground. To find the Gain in practice Kirchhoff's circuit laws can be used on the circuit on which the amplifier is connected to [16].

The bandwidth of an OP-Amp is a parameter that defines the range of frequencies over which the amplifier can effectively operate. A wider bandwidth generally implies better performance, allowing the amplifier to respond to a broader spectrum of input frequencies. If the bandwidth is wide there is a risk that noise also becomes amplified. When amplifying electrical signals is therefore important to take the bandwidth parameter of the OP-Amp into account as to not reduce the signal's SNR [16].

Another parameter regarding an OP-Amp's performance is it's slew rate. This parameter indicates how quickly the output voltage changes in reaction to variations in

the input signal. Normally this is measured in volts per microsecond, V/µs. In circuits that measure rapid changes, it is essential to have a fast slew rate since limitations could lead to distortion of the signal [16].

2.4.2 Schmitt Trigger

A Schmitt trigger is a circuit configuration that converts an electrical signal to a digital signal. This is done by a switch that turns on when the input voltage reaches above or below a certain threshold which sets the output to either the power supply voltage or 0. A Schmitt trigger is usually made up of components such as amplifiers, transistors and integrated circuits [27]. Because of how a Schmitt trigger behaves, it is often used to speed up the response of a circuit [27].

A hex inverting Schmitt trigger works inversely to a regular Schmitt trigger. This means that it switches on when the input voltage is below the threshold and turns off when the input voltage crosses the threshold.

2.4.3 Decoupling Capacitors

Decoupling capacitors are capacitors connected between the power supply and ground when using integrated circuits. They are used to remove high frequency noise from the power supply by effectively working as a short circuit for very high frequency signals [28]. These capacitors are vital to circuit construction when using many different integrated circuits such as Schmitt triggers and OP-Amps.

Chapter 3

Methods

This chapter explains the project methods that were carried out to fulfill the project tasks and aims. The chapter is split into four parts explaining the method behind getting the ideal signal, creating the signal generator, building the readout circuit and lastly using the data post-processing code. The post-processing code can be found in Appendix A.

3.1 The Ideal Signal

Using the time response function for a low pass filter, the ideal signal that is expected according to theory can be calculated. The voltage response follows the following equation for a step function input [29]:

$$v_0(t) = V_{max}(1 - e^{-\omega_0 t}) \tag{3.1}$$

Where $v_0(t)$ is the output voltage of the filter, V_{max} is the max voltage of the system and $\omega_0 = \frac{1}{RC}$.

The resistor and capacitor values were chosen to be 138 Ω and 3.3 µF to get a rise time of τ_r = 1 µs. The voltage response function and it's derivative were then plotted for analysis and the rise time and FWHM of the signal was calculated.

3.2 Signal Generator

A Hex inverting Schmitt trigger, 74HCT14N manufactured by Phillips, with a 6 ns rise time was connected to a breadboard. Feeding into the integrated circuit was a portable 5 V power supply and a function generator supplying a square wave signal with V_{pp} = 2 V, V_{offset} = 1 V and f = 3 Hz. The frequency f of the signal was chosen as 3 Hz to mimic the rotating disk with a rotational speed of 1 m/s. The output of the integrated circuit was connected to a low pass filter with a rise time of τ_r = 1 µs to achieve the correctly mimic the rise of the signal that the Faraday cup would generate. The OP-Amp OPA727AIDGKT, manufactured by Texas Instruments, was connected to the output of the low pass filter, configured according to Figure 2.2.1. A large resistor of 460 $k\Omega$ was connected after the active filter to limit the output current from the signal generator to 10 µA. The Schmitt trigger and the capacitor in the low pass filter was then connected to ground. Three decoupling capacitors of the magnitudes 47 nF, 220 nF and 10 µF were connected in parallel between the integrated circuits power supply pin and the ground pin. A Fourier analysis was also performed on the output of the signal generator to find out which frequencies are present in the signal.

To measure the signal straight from the signal generator, an oscilloscope probe was placed across the 460 $k\Omega$ to ground and 50 measurements were performed, overlapping 100 different waveforms and then averaging over them. A FIR filter was applied onto the averaged signal and the signal and it's derivative were plotted. The plots for all 50 measurements were saved and the signals rise time was calculated for each measurement. The derivatives FWHM was also calculated for all 50 measurements. How the overlapping and filtering was performed is further discussed in section 3.4

3.3 Readout Circuit

A 1 $k\Omega$ resistor was connected between the large resistor (460 $k\Omega$) in the signal generator and ground. An OP-Amp OPA727AIDGKT was connected to the output of the large resistor, configured as a non inverting amplifier with gain 50. A 50 Ω resistor was then connected from the amplifiers output to ground. A voltage inverting integrated circuit, ICL7660, was attached to the breadboard to supply both OP-Amps in the circuit with -5 V. The power inverting integrated circuit was supplied 5 V from

the portable power supply connected to the breadboard. The other pins on the power inverting integrated circuit were connected according to the standard configuration specified in the ICL7660 data sheet [30]. Lastly, decoupling capacitors were connected in parallel over both the positive and negative power pins of the OP-Amps as well as between the -5 V output of the inverting circuit and ground. Figure 3.3.1 shows the complete circuit including both the signal generator and the readout circuit without the decoupling capacitors.

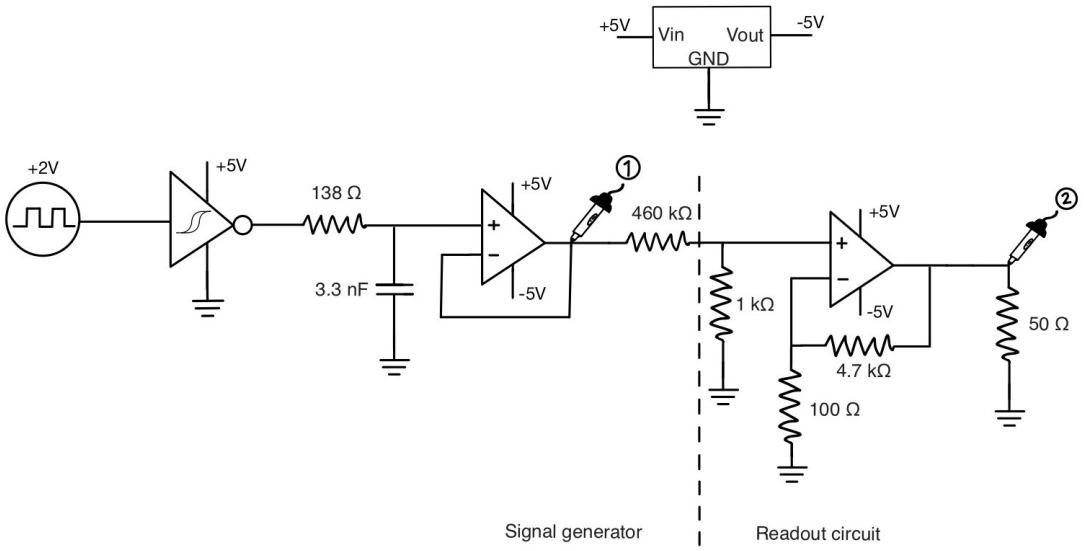


Figure 3.3.1: Circuit layout for the signal generator and readout circuit without decoupling capacitors. The probe marked 1 was used to measure the generated signal and the probe marked 2 was to measure the readout circuit signal.

3.4 Automatic Signal Measurement Code and Signal Measurements

Code for automatic data collection was written in Python using the Picoscope SDK library to allow communication with the portable PicoScope 2204A to automatically set and perform measurements. The PicoScope SDK (10.7.25) library, called picosdk.ps2000, was imported into a Python file in PyCharm 2022.2.3.

A $measure$ function was created which utilises the functions in the Picoscope SDK library to communicate and configure the picoscope to perform automatic measurements. The $measure$ function code connects to an available Picsoscope 2204A

and makes it perform a set amount of measurements by sending out a specified signal through the AWG channel and measuring the outputted signal on the in going channel. This function also sets the oscilloscope trigger to correctly identify the slope, which is the region of interest of the signal. The $measure$ function is then called in the Python codes main function multiple times and for multiple iterations to be able to overlap several signals and then average over them to mitigate noise and perform multiple measurements back to back of the same signal. The function $FIR\ filter$ takes signal data from $measure$ as input and passes it through a FIR filter to further filter and mitigate noise in the signal. The FIR filter used was the $signal.firwin$ function from the scipy library. The parameters used for the FIR filter were a cutoff frequency of 3 MHz and a blackman window function. The number of taps used was an odd number that was either set to $\frac{N}{2}$ or $\frac{N}{2} + 1$, where N is the number of samples taken, depending on whether $\frac{N}{2}$ was an even or odd number. The function $signal\ process$ takes the FIR filtered data as input and outputs that signals rise time, time constant, SNR and FWHM. These metrics were saved in a CSV file for further data analysis. A function called $calc\,radius$ takes the FIR-filtered signal as input and calculates the derivative of the signal using the central difference method. Lastly the code plots the raw signal, the averaged signal, the FIR-filtered signal and the derivative of the signal and saves the created plots.

To test the readout circuit the currents 10 µA, 5 µA and 2.5 µA, were used to see how the accuracy of the circuit changed with decreasing current magnitude. The change in current was achieved by increasing the resistance of the large resistor in the signal generator. 460 kΩ, 920 kΩ and 1.84 MΩ was used to test for 10 µA, 5 µA and 2.5 µA respectively. When testing the circuit at the three different currents the code collected 50 measurements, overlapping 100 waveforms of the signal. This was done by placing an oscilloscope probe across the 50 Ω resistor, seen in Figure 3.3.1 to ground.

Finally the data collected the CSV file was plotted onto three different histograms, one for SNR, one for Rise Time and one for FWHM. The mean rise time, SNR and FWHM was calculated for each current value from the data saved to the CSV file. The mean absolute difference between the signal generators FWHM and rise time was also calculated for each current value.

Chapter 4

Results

This chapter contains the results achieved through the project method. It is worth to note that Ref or reference refers to the signal from the signal generator. It is also worth to note that all plots denoted as ”derivative plots” have $\frac{dI}{dt}$ on the y-axis with the units mA/ns and x on the x-axis with the unis nm.

4.1 Ideal Signal

The ideal signal that the signal generator is expected to produce according to theory is shown in Figure 4.1.1 where voltage and time is on the y-axis and x-axis.

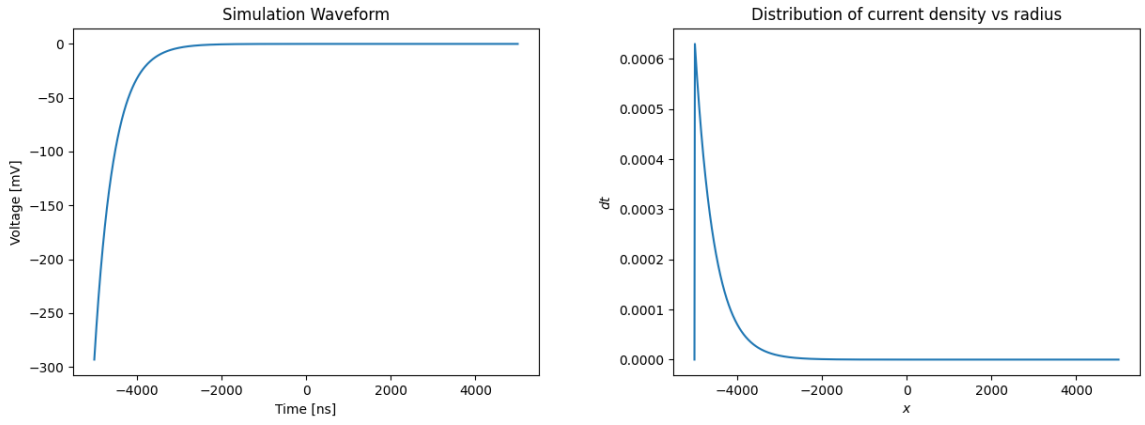


Figure 4.1.1: Ideal signal (left) and the derivative of the ideal signal (right)

Table 4.1.1 contains the calculated rise time and FWHM for the ideal signal shown in Figure 4.1.1.

Table 4.1.1: The rise time and FWHM for the ideal signal

Rise Time [ns]	FWHM [nm]
1000	310

4.2 Signal Generator

The raw signal and the FIR-filtered signal measured over the large resistor in the signal generator is shown in Figure 4.2.1 where voltage is on the y-axis and time is on the x-axis.

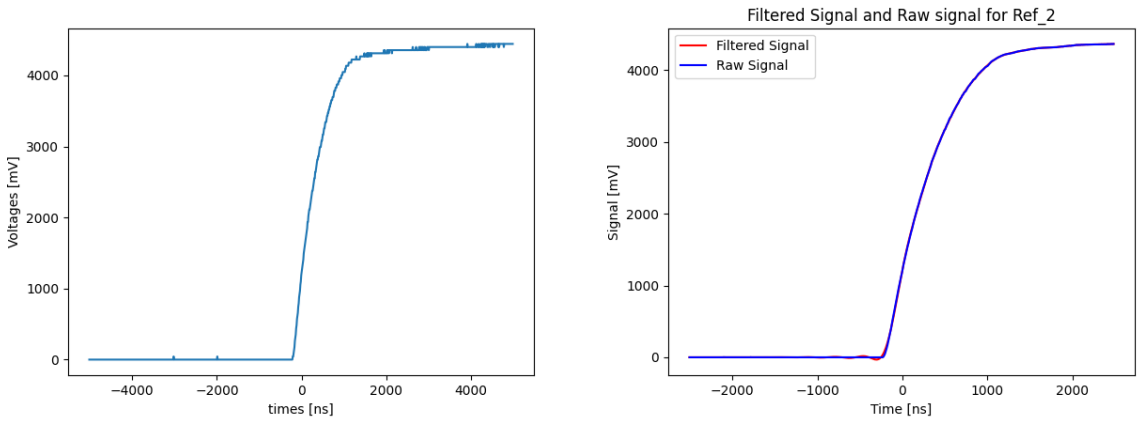


Figure 4.2.1: Raw signal from the signal generator (left) and the FIR-filtered signal from the signal generator (right)

The derivative of the signal generators signal is shown in Figure 4.2.2 where the slope is on the y-axis and signal position is on the x-axis.

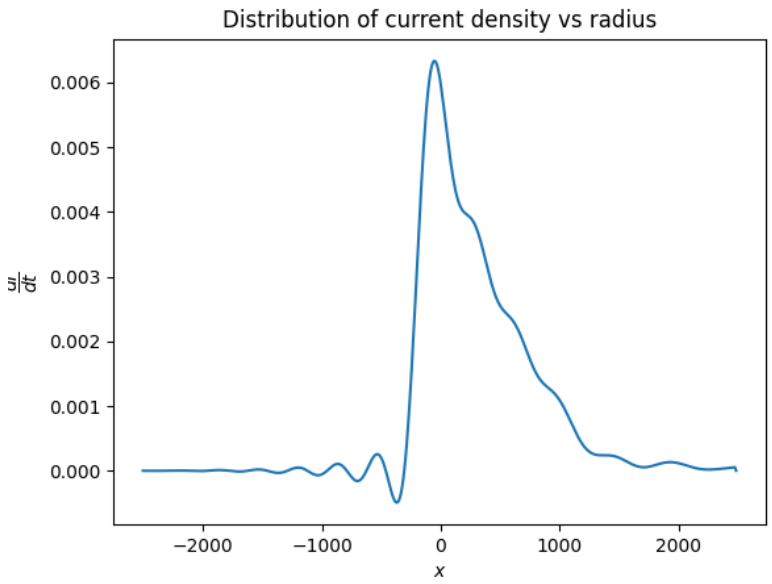


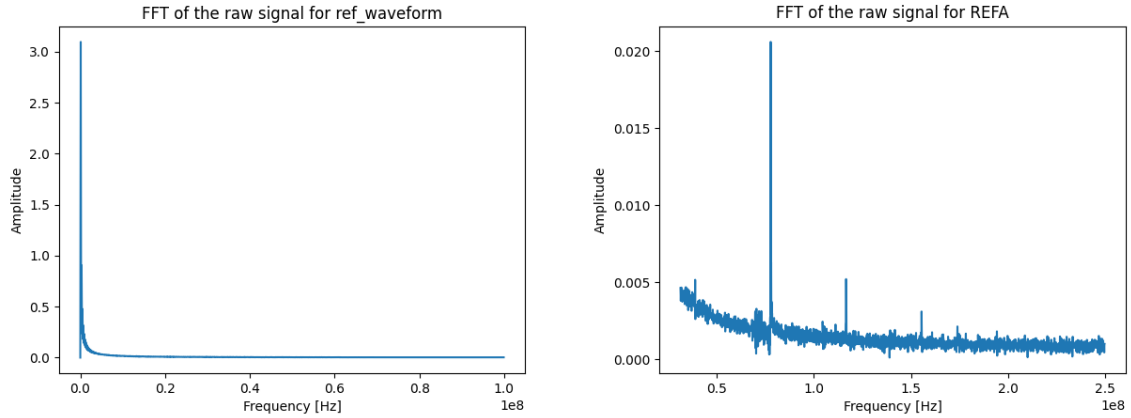
Figure 4.2.2: The derivative of the signal generators signal. The y-axis has values for $\frac{dI}{dt}$ in mA/ns and the x-axis is the width in nm

Table 4.2.1 shows the average rise time, SNR and FWHM for the FIR-filtered signal measured over the large resistor in the signal generator.

Table 4.2.1: The average rise time, SNR and FWHM for the FIR-filtered signal measured over the large resistor in the signal generator.

Rise Time [ns]	SNR	FWHM [nm]
1010.2	4630.7	579

The figure below shows the results from the Fourier analysis performed on the signal in Figure 4.2.1.



Full FFT curve Zoom in of the Full FFT curve

Figure 4.2.3: Fourier analysis results from the signal output of the signal generator

4.3 Readout Circuit Measurements

Figure 4.3.1, 4.3.2 and 4.3.3 shows both the FIR-filtered signal and the derivative of the signal measured over the 50 Ω resistor in the readout circuit to ground for 10 µA, 5 µA and 2.5 µA. The measured current from each were 9.6 µA, 5.2 µA and 2.5 µA respectively.

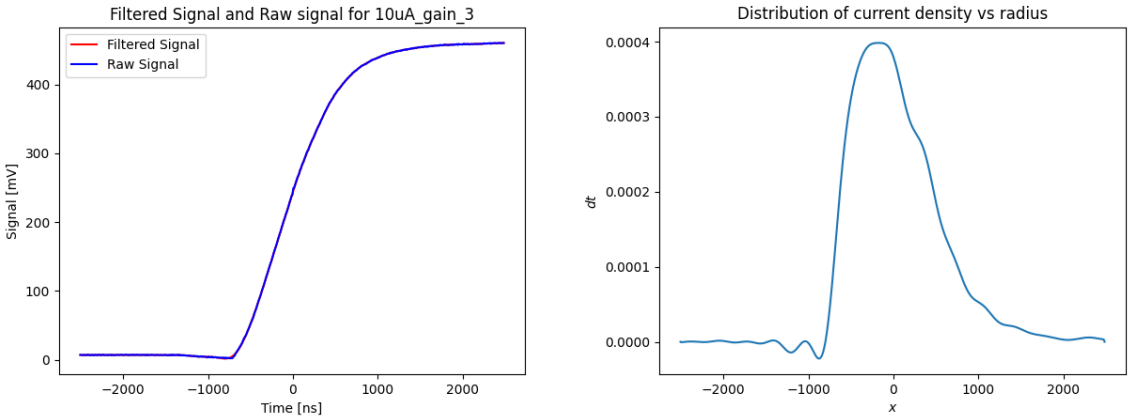


Figure 4.3.1: FIR-filtered signal (left) and the derivative of the signal (right) measured over the 50 Ω resistor in the readout circuit for 10 µA.

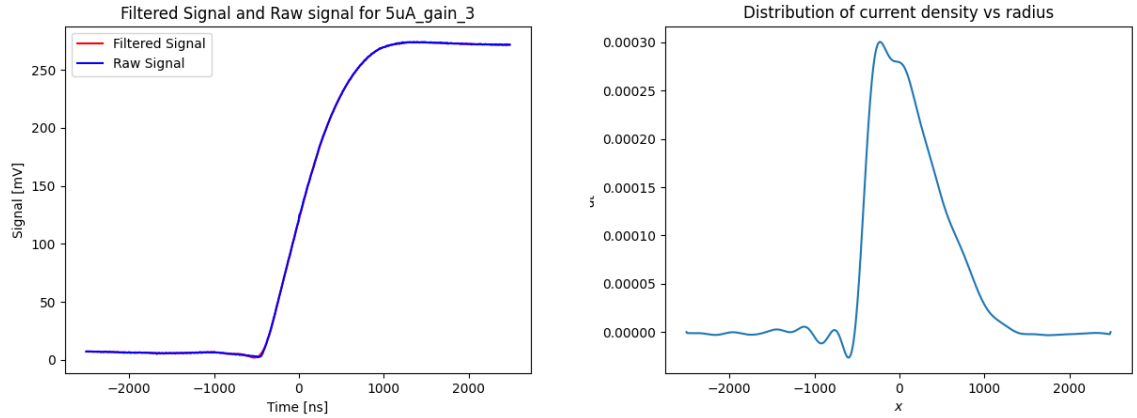


Figure 4.3.2: FIR-filtered signal (left) and the derivative of the signal (right) measured over the 50 Ω resistor in the readout circuit for 5 µA.

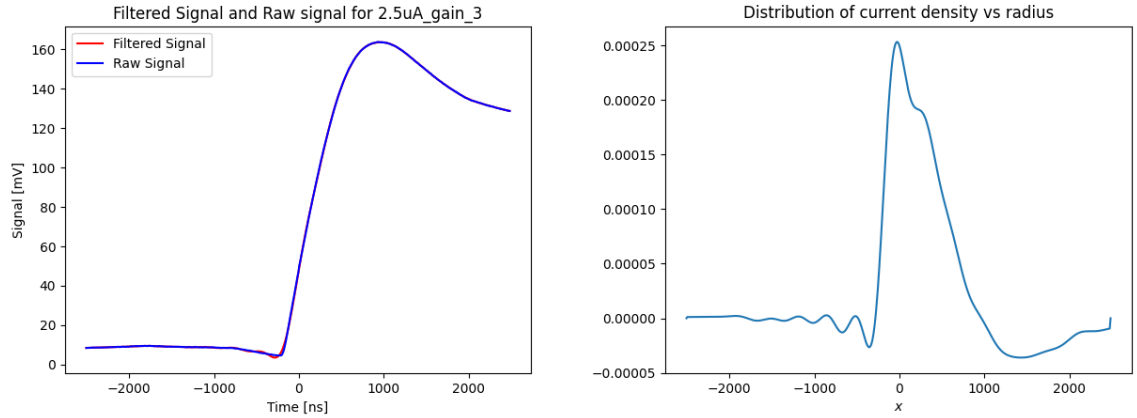


Figure 4.3.3: FIR-filtered signal (left) and the derivative of the signal (right) measured over the 50 Ω resistor in the readout circuit for 2.5 µA.

The average rise time, SNR and FWHM for the signal measured over the 50 Ω resistor in the readout circuit for 10 µA, 5 µA and 2.5 µA is shown in Table 4.3.1.

Table 4.3.1: The average rise time, time constant, SNR and FWHM for the signals measure over the 50 Ω resistor in the readout circuit for different current magnitudes.

Current [µA]	Rise Time [ns]	SNR	FWHM [nm]
10	1198.8	1743.5	1101.6
5	956.8	939.9	872.2
2.5	682.4	536	640.2

4.4 Error Estimation and Statistics

Figure 4.4.1 is a histogram over the SNR values for the different measurements performed above.

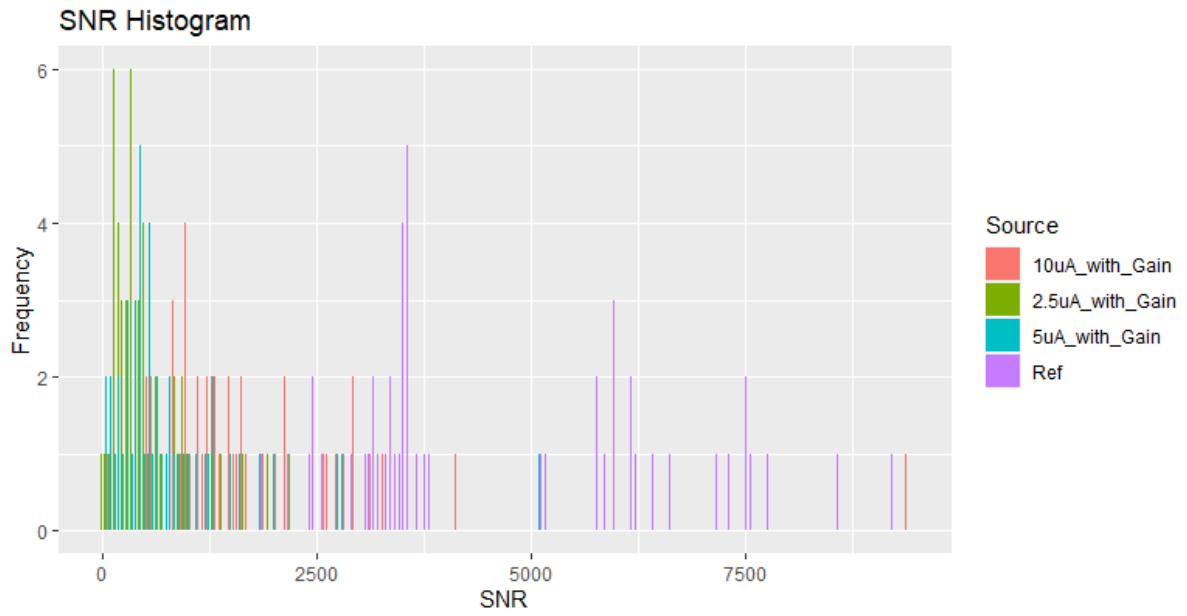


Figure 4.4.1: Histogram of the SNR values for the different measurement series

Figure 4.4.2 is a histogram over the Rise Time values for the different measurements performed.

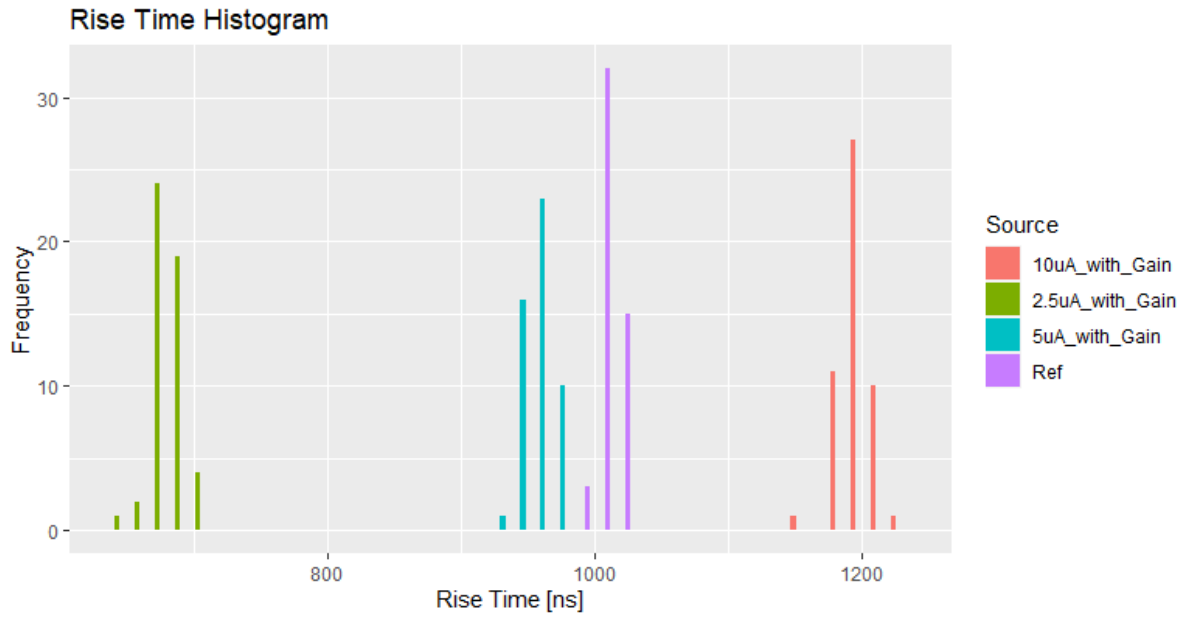


Figure 4.4.2: Histogram of the Rise Time for the different measurement series

Figure 4.4.3 is a histogram for the FWHM values for the different measurements performed.

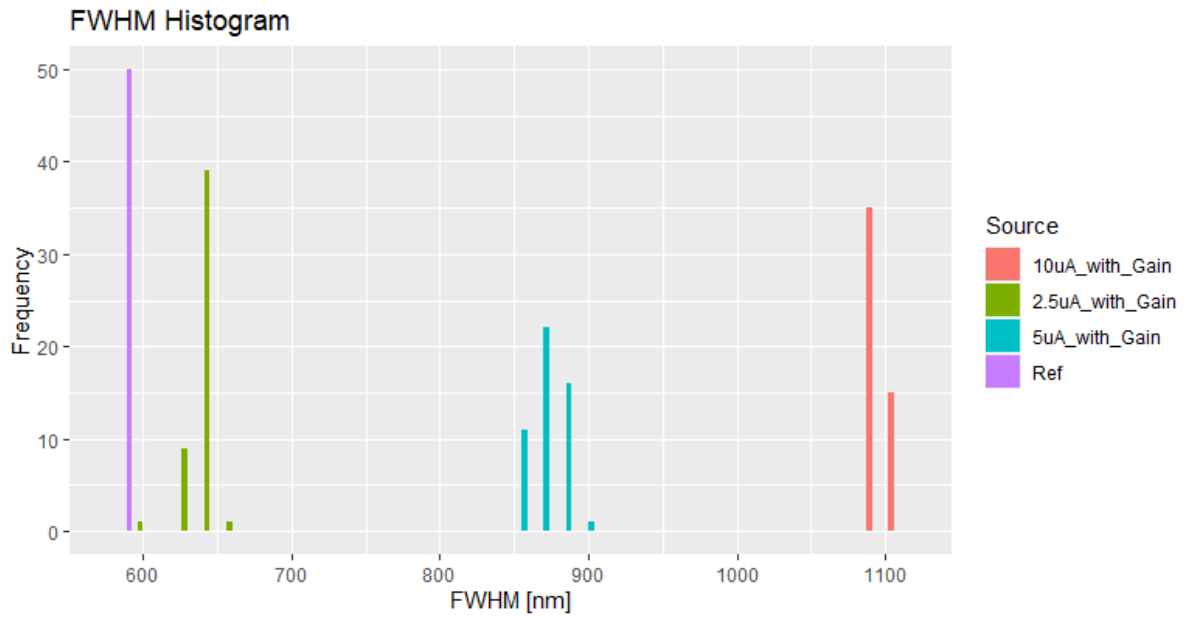


Figure 4.4.3: Histogram of the FWHM values for the different measurements

The table 4.4.1 below contains the mean absolute difference for the Rise time and the FWHM between each current measurement and the reference signal (signal from the signal generator).

Table 4.4.1: The mean absolute differences for the Rise time and FWHM between each current measurement and the reference signal

Current [µA]	FWHM [nm]	Rise Time [ns]
10	522.6	188.6
5	293.2	53.4
2.5	61.2	327.8

Chapter 5

Discussion

This chapter discusses and evaluates the results achieved through the project's methods. The discussion of the results follow in the same order as they are presented in the results chapter. Furthermore, this chapter discusses the environmental and ethical aspects revolving the project. One thing that is important to note is that all of the signal plots in the result have mV on the y-axis. To find the current of the beam from this signal one has to simply divide the signal in volts by 50000 to account for the applied gain to the signal and the voltage dividing effect provided by the 1 $k\Omega$ resistor before the amplification step.

5.1 The Ideal Signal

The theoretical signal that the signal generator is supposed to emulate is shown in Figure 4.1.1 along with its respective derivative. The ideal signal is derived from the time response function for a low pass filter shown in Equation 3.1. The resistor and capacitor values were chosen in such a way that the time constant was 1 µs. Calculating the rise time of the signal shown in Figure 4.1.1 it is confirmed that it is exactly 1 µs and with a FWHM of 310 nm, shown in Table 4.1.1. These results acts as a reference of what was theoretically expected from the circuit designed in the project.

5.2 The Signal Generator

The Schmitt triggers 6 ns rise time was intentionally chosen as to make the circuit faster than required as to later slow it down. By connecting a low pass filter this makes

it possible to limit the output signals rise time to the desired value of 1 µs. The produced signal, raw and filtered, can be seen in Figure 4.2.1 where they appear similar to the ideal signal. It is worth to note that measurements taken from the signal generator do not show the limiting of the current rather it only shows how the waveform should ideally look like. Calculating the rise time for the filtered signal we get a slightly longer rise time of 1.01 µs and a larger FWHM of 579 nm, shown in Table 4.2.1. Looking at the derivative of the filtered signal seen in Figure 4.2.2 and comparing it to the derivative of the ideal signal we can identify two artifacts. The first one is the oscillations right before the sharp increase and the second one is the curves ring down effect after the maximum. The first artifact does not create any issues because when calculating the rise time and FWHM that part of the signal is not used. However, the second artifact affects the decent of the derivative curve which in turn affects the FWHM value, making this value larger than it should be.

The first artifact is an effect of the FIR filter which generates ringing at points where there are changes in the curve much quicker than the cutoff frequency. The FIR filter was employed as even though the signal is overlapped and averaged 100 times, a small amount of high frequency noise (with low amplitude) makes it through. This causes the derivative of the signal to become very noisy as the rapid change in slope leads to an ever larger oscillation in the derivative space. This would make calculating the FWHM algorithmic-ally extremely tedious and complicated.

The second artifact could arise from two potential factors, signal reflections and/or aliasing. The OP-Amp chosen for the active filter (OPA727AIDGK) was primarily chosen as it has a fast enough slew-rate for the desired application of 30 V/µs, but it was also chosen as it has a bandwidth of 20 MHz. Figure 4.2.3 shows that the desired signal is mainly composed of frequencies under 20 MHz, with all the harmonic effects (presumably from the changes in the slope) falling off at around 16 MHz. This means that any high frequency noise that may pass through the filter should be further filtered out by the OP Amp. The active filter was chosen as to separate the separate parts of the circuit as to reduce the chance of reflections from occurring. That being said, reflections can still occur at the point of measurement as the resistors used to reduce the current had a size of 460 kΩ, 920 kΩ and 1.84 MΩ and the oscilloscope probe has an inner resistance of 50 Ω [24]. This is a potential explanation of the ringing seen in Figure 4.2.2 and can be solved by further modifying the total impedance of the circuit to be matched to 50 Ω.

Even though aliasing is extremely unlikely, due to the cutoff frequency and the bandwidth of the OP-Amp being significantly below 50 MHz (Nyquist freqency), the slope may have small sections of change that are above this threshold. This could cause small distortions in the slope which may be what is being seen in Figure 4.2.2. To solve this problem one could use an oscilloscope with a significantly higher sampling rate than 100 MHz.

5.3 Readout Circuit Measurements

By analyzing Figures 4.3.1 - 4.3.3 it can be seen that reducing the current causes the readout circuit to have an increasing overshoot. According to the datasheet for the OP-Amp, when amplifying small signals, if there is a lack of a capacitive filter the OP-Amp will generate a slight overshoot [31] (Figure 26 in the datasheet). This explains the phenomenon seen in Figure 4.3.2 and 4.3.3 and can be solved by adding that small capacitance of a few picofarads to the feedback loop. Even though this may solve the problem it could also distort the signal by working as an active low pass filter. This trade-off can be found by trial and error and implemented in the readout circuit when used with the real Faraday cup. It can also be seen that the ringing in the derivative plots seen in Figure 4.2.2 is less prominent in the readout circuit results. This could mean that there are less reflection artifacts as the impedance along the readout circuit better matches the impedance in the oscilloscope probe.

It can also be seen that the current values for the performed measurements from the circuit are slightly off for 10 µA and 5 µA from their true values. This is due to the signal generator not having exactly the correct resistor to reduce the current to exactly the correct value when calculating the expected voltage drop using Ohm's law. It is also due to small parasitic impedance values that cause a voltage drop in the signal, causing these values to be slightly incorrect. This being said, these are errors generated primarily by the signal generator which means that the readout circuit accurately measures the current of the signal coming into it, limited only by the resolution of the oscilloscope. The resolution of the oscilloscope is 12 bits [25] which can then be converted to decimal points of accuracy using the following equation [32]:

$$\text{Decimal Points of Accuracy} = log_{10}(2^{\text{Number of Bits}}) \tag{5.1}$$

Which for 12 bits equates to roughly 3 decimal points of accuracy. This means that the measured current is accurate for up to 0.0001 mV given that the signal is expected to be measured in mV.

Table 4.3.1 shows that as the current is decreased, the SNR value of the circuit also decreases. It also shows that the rise time calculated and the FWHM values decrease. The SNR decreasing makes sense as the smaller the input signal, the more prominent the low-amplitude noise is in the signal. This being said the inconsistency in rise time and FWHM do not make sense from a purely theoretical perspective and are due to some artifact present in the measurement. The rise time for the measurement of 2.5 μA being highly different from the other two can be explained as due to the previously mentioned overshoot, which causes the circuit to respond much quicker than for the other current values.

The inconsistencies in FWHM, on the other hand, depend on something entirely different. Looking at the derivative plots in Figures 4.3.1-4.3.3 it can be seen that around $x = 0$, the signal seems to flatten out at the maximum value. This artifact probably occurred due to one of the following factors; a sampling error, an overlapping error or a power-supply error.

Due to the sampling rate being capped at 100 MHz it is possible that small changes in the slope are missed due to under-sampling which could potentially cause some sort of aliasing. This can be tested by analyzing the signal with an oscilloscope that has a much higher sampling rate.

As the signals are being overlapped digitally using the triggering mechanism of the oscilloscope it is possible that there is a misalignment in the triggering of a few ns. This could potentially explain why the change in slope that is expected at $x = 0$ is missed as the program overlaps the signal slightly incorrectly. To account for this error one could use an oscilloscope with a more accurate trigger or manually align the plots in software. This would also be easier to accomplish if the oscilloscope used had a higher sampling rate.

Since a voltage inverting integrated circuit was used to provide the OP-Amps with -5 V it is possible that the performance of the OP-Amp is slightly depreciated. This is because the integrated circuit is not always stable and is very sensitive to small changes in the power supply and impedance in the circuit. This drop in performance could potentially cause the plateauing of the derivative but it can also be a potential factor

that causes the ringing effect seen in all of the derivative plots. To solve this problem one could instead use batteries connected in series. This should eliminate errors that arise from instabilities in the voltage inverting integrated circuit.

5.4 Error Estimation and Statistics

The results in Figure 4.4.1 show that lowering the current tends to decrease the SNR. This being said as long as the SNR value is greater than 1, it should be possible to distinguish the signal from noise. The figure shows that the bulk of the measurements have an SNR value around 1000 which means that the signal is strong enough in most cases to distinguish it from the background noise.

Figure 4.4.2 shows that the 5 μA measurements yielded results that were closest to the signal from the signal generator. This was followed by the measurements from the 10 μA signal and finally the 2.5 μA signal that gave a rise time furthest away from the reference value. Table 4.4.1 further confirms these results. This makes sense due to the artifacts and characteristics of the signal discussed in section 5.3.

Figure 4.4.3 on the other hand shows that 2.5 μA measurement is the closest to the reference while increasing the current seems to make the FWHM further away from the reference. Table 4.4.1 further confirms these results. This makes sense as in section 5.3 it is seen that there is an artifact in the signal that causes the derivative to plateau at around $x = 0$, which causes the FWHM to be larger than the ideal situation. For the 2.5 μA measurement though, the overshoot causes the signal to reach this point much quicker which in turn makes the FWHM closer to the reference signal.

The histograms in Figure 4.4.2 and Figure 4.4.3 show that for each measurement, the spread of measured values are relatively small which shows that the readout circuit has a relatively high precision. One could also perform measurements for current values inbetween 2.5 μA and 10 μA to find out the trend of the bias that the readout circuit seemingly has and then use this information to generate a regression model that can compensate for the systematic error for that specific current.

5.5 Readout Circuits Performance in the real Experimental Setup

It is generally hard to estimate how the circuit will perform in the real experimental setup as the Faraday Cup to be used is not yet constructed and it's electronic characteristics have not yet been well defined. That being said the reflection artifacts are expected to go down as there should be less parasitic impedance and it should therefore make it easier to match the impedance of the load to the rest of the circuit. One could also reduce the effects of reflection by using a well designed PCB for the circuit as to reduce parasitic inductance from wires and parasitic capacitance from breadboard connectors.

By following the recommendations in the previous sections by changing the power supply, increasing the sampling rate by using an oscilloscope with a better temporal resolution and by tweaking the impedance of the circuit one should be able to remove a large amount of the artifacts present in the signal which should ideally generate an accurate measurement. Since in the experimental setup, the signal generator will not be present, a large amount of noise and distortions generated by the Schmitt trigger and active filter components will be removed and therefore the signal measured should have smaller systematic errors. It is also expected that the readout circuit will accurately measure the beam current during the experiment.

5.6 Ethical Considerations

Engineers have a responsibility to not cause harm with their inventions and devices. In this project, a lot of electrical components were used. One of the components used in this project is OPA727AIDGK which is produced by Texas Instrument. On their website they discuss their work toward good and sustainable relationships between their investors, suppliers and employees. They believe in treating them all with respect and dignity. Texas Instruments claim that they uphold human rights and ethical practices. They also expect the same commitment from their partners and suppliers, this by being a member of Responsible Business Alliance (RBA) [33]. The foundation was founded in 2004 by a group of leading electronics companies. RBA was created for the purpose of corporate social responsibility in global supply chains. All RBA members must ”utilize a range of RBA training and assessment tools to support

continuous improvement in the social, environmental and ethical responsibility of their supply chains"[34]. All RBA members must also commit to follow the common Code of Conduct [34].

The company selling the Schmitt trigger is Phillips, but the actual manufacturing is being done by the company NXP Semiconductors. A new chapter that took place within the company in 2022, is when the company released their first Human Rights Policy. It reads that "We are committed to business practices and technologies that contribute to protecting and respecting human rights worldwide. NXP is therefore committed to supporting and upholding all international labour and human rights laws and norms" [35].

This means that the components used in the project are ethically produced and it is therefore ethically justified to incorporate the products used in this specific project. The circuit designed in the project will also not be used to harm anyone and was solely built for research purposes that will hopefully help innovate in the field of electron microscopy.

5.7 Environmental Impact of the Components Used

The project used a lot of electrical components such as resistors, capacitors, wires, solder and Schmitt triggers. It is important that these components are correctly recycled and/or reused as such to not cause a large amount of e-waste. It is also important to consider that these components are constructed according to the correct regulations and that their environmental effects are taken into account.

Texas Instruments [36] discuss about their work towards a more sustainable manufacturing. This includes reducing their energy consumption, thus reducing greenhouse gas emission, by using alternative energy sources during production. This is accomplished, according to them, by upgrading their factory tools and abatement technology. They also have the ambition to reuse and recycle significantly more of their waste, surplus materials and water.

Regarding sustainability, NXP writes in the company's Corporate Sustainability Report 2022 [37] that they have significantly reduced their emission which is still a continuous

goal every year. They have also changed energy sources into renewable electricity. The company has many sustainable long term goals, such as being carbon neutral in 2035 and they plan to develop collaborative circular economy solutions for their products [38]. Circular economy embraces a model that aims at prolonging products life, reducing waste, and recycling materials to be used again [39]. According to the company's 2020 sustainability policy they are working towards creating new products that aim at reducing the environmental impact in the sense of how the product is used and how it will be disposed of [40].

This implies that the components used in the project were manufactured with environmental considerations taken into account. The excess components for the project will be handed over to KTH to dispose of or to reuse. The main circuit designed will be handed over to the team that will continue with the project.

5.8 Alternative and Tested Solutions

Before coming to the final design a few other alternative solutions were tested for both the readout circuit and the signal generator respectively. In this section these solutions and other possible solutions other than the one presented in this report will be discussed.

5.8.1 Signal Generator

Before arriving at the signal generator presented in this report, a signal generator was created using only a passive low pass filter. This approach did not work for the intended use as the rise time of this generator was heavily affected by other impedance sources in the circuit (Schmitt trigger and current limiting resistor). The low pass filter's performance changed and caused the rise time of the circuit to be significantly larger than the calculated value. On top of this, the impedance of the circuit was significantly larger than the load impedance causing more reflection artifacts when measuring the signal.

An alternative solution that was not tested in this project would be to use a pre-built signal generator that already has a rise time of 1 μs. This could eliminate artifacts and noise generated by the signal generator proposed in this report.

5.8.2 Readout Circuit

The original approach was to only use a single resistor to measure the incoming current. While this in theory works, due to the reflections coming from the current limiting resistor connected in series to this resistor, a large amount of the signal was reflected back causing both the signal and rise time to be heavily distorted when using large resistor values. When using small resistor values the signal was extremely small and very susceptible to noise which made it hard for the trigger to correctly identify the slope and trigger correctly. This in turn made filtering and overlapping the signal a tedious and inaccurate process.

When measuring small currents that need to be resolved in time when using a Faraday cup there are three different methods that could be used. The first two are the methods employed in this project, namely using a resistor or an amplifying circuit. The third method, which was not tested in this project, is to use an induction loop to measure the generated current. An induction loop is a coil that, when alternating current passes perpendicularly through it, generates a current [41]. This works by using the following of Maxwell's equations:

$$\nabla \times \vec{B} = \mu_0(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \tag{5.2}$$

Where $\vec{B}$ is the magnetic field, μ_0 is the permeability in vacuum, $\vec{J}$ is the current density, ϵ_0 is the permittivity in vacuum and $\frac{\partial \vec{E}}{\partial t}$ is the change of the electric field in time. It implies that the change in current density (when current comes in through the Faraday cup) produces a magnetic field which in turn induces a current in the nearby loop using the Maxwell-Faraday equation, which is stated as follows:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{5.3}$$

This means that it is in theory possible to measure currents with the help of an induction loop that produces an electric field which in turn produces a voltage which then finally produces a current. That being said it is unclear how this would perform with the proposed experimental setup. A primary reason as to why this was not tested was as there is not a lot of information regarding how this is done in practice and what type circuit is required to correctly process the signal without major distortions. It is also unclear without further calculations how the measured signal behaves with

regards to the original signal from the Faraday cup.

If the circuit proposed in this report, with the improvement suggestions, were to not perform well with the real life Faraday cup it would be worth to test both the single resistor setup and the induction loop respectively.

5.9 Contributions

This section will talk about the contributions made by the different authors of the report. It highlights who contributed to which part of the various different parts of the project and report.

5.9.1 Smyan

Smyan helped in designing, creating and testing the circuit, writing the post processing code, data analysis and data collection. He also wrote parts of the introduction, background, method, results, and large parts of the discussion and conclusion.

5.9.2 Jingyu

Jingyu helped in designing and creating the circuit. He also wrote parts of the background relating to the Schmitt trigger, oscilloscope and experimental setup.

5.9.3 Anton

Anton helped in designing and creating the circuit. He also wrote parts of the background, method and discussion. He also helped making the poster.

5.9.4 Hugo

Hugo helped in designing, creating and testing the circuit and data collection. He also wrote parts of the abstract, introduction, background, method, results and discussion.

5.9.5 Elin

Elin helped in designing and creating the circuit. She also wrote parts of the abstract, introduction and discussion.

Chapter 6

Conclusion

In conclusion, it is possible to measure 2.5 μA and upwards of current with varying accuracy with the circuit designed in this project. The closest FWHM to the reference signal was produced with the 2.5 μA signal and the closest rise time was produced with the 5 μA signal. The deviations from the reference signal measured during these measurements can be reduced by tweaking the readout circuit, by impedance matching and stabilizing the power supply, and by increasing the temporal resolution of the oscilloscope. It is also believed that the readout circuit will have less artifacts and distortions when used with the real experimental setup.

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Appendix A

Code for data acquisition and analysis

Below is the python3 code used to perform the data acquisition and the data analysis.

```
from picosdk.ps2000 import ps2000
from picosdk.PicoDeviceEnums import picoEnum
from picosdk.functions import assert_pico2000_ok, adc2mV
from ctypes import byref, c_int16, c_int32, c_byte, sizeof
from time import sleep
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq
from scipy import signal, stats
import csv

SAMPLES = 1000
OVERSAMPLING = 1

def get_timebase(device, wanted_time_interval):
    current_timebase = 1
    time_interval = c_int32(0)
    time_units = c_int16()
    max_samples = c_int32()
    while ps2000.ps2000_get_timebase(
```

```
            device.handle,
            current_timebase,
            SAMPLES,
            byref(time_interval),
            byref(time_units),
            OVERSAMPLING,
            byref(max_samples)) == 0 \
            or time_interval.value < wanted_time_interval.value:
        current_timebase += 1
        if current_timebase.bit_length() > sizeof(c_int16) * 8:
            raise Exception('No appropriate timebase was identifiable')
    return current_timebase - 1, current_timebase


def measure(n, v_range, name):
    with ps2000.open_unit() as device:
        print('Device Info: {}'.format(device.info))

        res = ps2000.ps2000_set_sig_gen_built_in(
            device.handle,
            1_000_000,
            2_000_000,
            1,
            3.18,
            3.18,
            0.0,
            0.1,
            0,
            1_000
        )

        assert_pico2000_ok(res)

        sleep(0.1)

        res = ps2000.ps2000_set_channel(device.handle,
                                        picoEnum.PICO_CHANNEL['PICO_CHANNEL_B'],
```

```
                                    False,
                                    picoEnum.PICO_COUPLING['PICO_DC'],
                                    v_range)

    assert_pico2000_ok(res)

    res = ps2000.ps2000_set_channel(device.handle,
                                    picoEnum.PICO_CHANNEL['PICO_CHANNEL_A'],
                                    True,
                                    picoEnum.PICO_COUPLING['PICO_DC'],
                                    v_range)

    assert_pico2000_ok(res)

    #Change to 4096 when measuring over 1k ohm resistor
    #Change to 8192 when measuring direct
    #Change to 12288 when measuring over op amp for 10uA current
    #This value depends on both the window range
    #and the expected signal value.
    res = ps2000.ps2000_set_trigger(
        device.handle,
        picoEnum.PICO_CHANNEL['PICO_CHANNEL_A'],
        c_int16(8192), #This value should be chosen between +32767 and
                       #-32767 where the largest number is the max
                       #voltage of the chosen window size.
        0,
        -50,
        0
    )

    assert_pico2000_ok(res)

    timebase_a, interval = get_timebase(device, c_int32(10))


    collection_time = c_int32()
```

```
    def run():
        res = ps2000.ps2000_run_block(device.handle,
                                      SAMPLES,
                                      timebase_a,
                                      OVERSAMPLING,
                                      byref(collection_time))

        assert_pico2000_ok(res)

        while ps2000.ps2000_ready(device.handle) == 0:
            sleep(0.1)

        times = (c_int32 * SAMPLES)()

        buffer_a = (c_int16 * SAMPLES)()

        overflow = c_byte(0)

        res = ps2000.ps2000_get_times_and_values(
            device.handle,
            byref(times),
            buffer_a,
            None,
            None,
            None,
            byref(overflow),
            2,
            SAMPLES
        )

        assert_pico2000_ok(res)
        channel_a_overflow = (overflow.value & 0b0000_0001) != 0

        ps2000.ps2000_stop(device.handle)

        channel_a_mV = adc2mV(buffer_a,
                              v_range,
```

```
                                c_int16(32767))

    return list(times), list(channel_a_mV)

times, voltages = run()

raw_fig = plt.figure()
plt.plot(times, voltages)
plt.xlabel("times [ns]")
plt.ylabel("Voltages [mV]")
raw_fig.savefig(f"Raw waveforms 2.5uA gain\\Raw_{name}.png")
plt.close(raw_fig)

labels = ['Times (ns)', 'Voltages (mV)']

time_arr = []
V_arr = []
for i in range(n):
    time_val, V_val = run()
    if V_val[0] < V_val[-1]:
        time_arr.append(time_val)
        V_arr.append(V_val)

final_t = []
final_V = []
vals = np.linspace(0, 0, num=SAMPLES)
for i in range(len(time_arr)):
    vals = np.add(vals, time_arr[i])

final_t = [x/n for x in vals]
vals = np.linspace(0, 0, num=SAMPLES)
for i in range(len(V_arr)):
    vals = np.add(vals, V_arr[i])

final_V = [x/n for x in vals]

float_fig = plt.figure()
```

```
        plt.plot(final_t, final_V)
        plt.xlabel("times [ns]")
        plt.ylabel("Voltages [mV]")
        float_fig.savefig(f"Float waveforms 2.5uA gain\\Float_{name}.png")
        plt.close(float_fig)

        return final_t, final_V, voltages

def signal_process(times, S, V, SNR_val, FWHM):

    S_corr = np.subtract(S, np.min(S))

    max_amplitude = max(S_corr)
# Replace with peak-to-peak if the signal oscillates around zero

    low_ref = 0.1 * max_amplitude
    high_ref = 0.9 * max_amplitude

    t_low = None
    t_high = None

    for time, value in zip(times, S_corr):
        if t_low is None and value >= low_ref:
            t_low = time
        if t_low is not None and value >= high_ref:
            t_high = time
            break

    rise_time = None
    tau = None

    if t_low is not None and t_high is not None:
        rise_time = t_high - t_low
        tau = rise_time / 2.2
        print("Rise Time:", rise_time, "[ns]")
        print("Tau = ", tau, "[ns]")
    else:
```

```
        print("Rise time couldn't be determined.")

    #SNR = (max(S) / np.max(np.subtract(S, V))) ** 2

    SNR = SNR_val

    print("SNR = ", SNR)

    params = [SNR, tau, rise_time, FWHM]

    with open("5uA_gain_data.csv", "a", newline='') as file:
        writer = csv.writer(file)
        writer.writerow(params)



def generate_fft(times, V, name):
    times = [x * 10 ** (-9) for x in times]
    V = [x * 10 ** (-3) for x in V]

    N = len(V)
    T = abs((times[1] - times[0]))

    print(1 / T)

    SNf = fft(np.subtract(V, np.mean(V)))
    xf = fftfreq(N, T)[:N // 2]

    fig1 = plt.figure()
    plt.plot(xf, 2.0 / N * np.abs(SNf[0:N // 2]))
    plt.xlabel("Frequency [Hz]")
    plt.ylabel("Amplitude")
    plt.title(f"FFT of the raw signal for {name}")
    fig1.savefig(f"FFT_of_{name}.png")

def FIR_filter(times, V, name, raw_V):
    times = [x * 10 ** (-9) for x in times]
```

```
V = [x * 10 ** (-3) for x in V]

N = len(V)
T = abs((times[1] - times[0]))

numtaps = int(N / 2) if (int(N / 2) % 2) != 0 else int(N / 2) + 1

h = signal.firwin(numtaps=numtaps, cutoff=3 * 10 ** 6, window="blackman",
                  fs=1 / T)

S = np.convolve(V, h, mode="valid")

trim_size = (numtaps - 1) // 2

# Adjust the V and times arrays to match the size of S
V = V[trim_size:-(trim_size or None)]
times = times[trim_size:-(trim_size or None)]
raw_V = raw_V[trim_size:-(trim_size or None)]


times = [x * 10 ** 9 for x in times]
V = [x * 10 ** 3 for x in V]
S = [x * 10 ** 3 for x in S]

SNR = (max(S) / np.max(np.subtract(S, raw_V))) ** 2

"""
fig2 = plt.figure()
plt.plot(times, S, color="r")
plt.xlabel("Time [s]")
plt.ylabel("Signal [V]")
plt.title(f"Filtered Signal for {name}")
fig2.savefig(f"Filter_Signal_{name}.png")
"""

fig3 = plt.figure()
plt.plot(times, S, color="r", label="Filtered Signal")
```

```
    plt.plot(times, V, color="b", label="Raw Signal")
    plt.xlabel("Time [ns]")
    plt.ylabel("Signal [mV]")
    plt.legend()
    plt.title(f"Filtered Signal and Raw signal for {name}")
    plt.show()
    fig3.savefig(f"FIR waveforms 2.5uA gain\\FIR_Raw_{name}.png")
    plt.close(fig3)


    return times, S, SNR

def calc_radius(S, resis, rotation, times):
    dI = np.zeros(len(S))
    current = np.divide(S, resis)
    distances = np.multiply(times, rotation)
    dt = times[1]-times[0]

    for i in range(len(S)):
        if 0 < i < len(S)-1:
            dI[i] = (current[i+1]-current[i-1])/(2*dt)


    return dI, distances


if __name__ == "__main__":
    choice = int(input("Choose Your Current: (1)5V (2)50mV (3)200mV (4)1V
                    (5)500mV: "))

    if choice == 1:
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_5V']
    elif choice == 2:
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_50MV']
    elif choice == 3:
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_200MV']
    elif choice == 4:
```

```
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_1V']
    elif choice == 5:
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_500MV']
    else:
        V_range = ps2000.PS2000_VOLTAGE_RANGE['PS2000_5V']


    nr_of_loops = 1
    for i in range(nr_of_loops):
        times, S, V = measure(100, v_range=V_range, name=f"2.5uA_gain_{i}")
        times, S, SNR = FIR_filter(times, S, name=f"2.5uA_gain_{i}", raw_V=V)

        dI, distances = calc_radius(S=S, resis=1000, rotation=1, times=times)

        # Assuming dI and distances are numpy arrays of the same length
        dI = np.array(dI)  # Replace with your intensity data
        distances = np.array(distances)

        # Below are the steps to calculate FWHM:
        # Step 1: Find the maximum intensity and its corresponding index
        max_intensity = np.max(dI)
        half_max = max_intensity / 2

        # Step 2 & 3: Find the points where the intensity crosses half the maxim
        # The approach here is to find the first point to the left of the
        # peak and the first point to the right of the peak
        left_index = np.where(dI >= half_max)[0][0]
        right_index = np.where(dI >= half_max)[0][-1]
        # Step 4: Calculate the FWHM
        FWHM = distances[right_index] - distances[left_index]

        print("FWHM:", FWHM)
        rad_fig = plt.figure()
        plt.plot(distances, dI)
        plt.ylabel(r"$\dfrac{dI}{dt}$")
        plt.xlabel(r"$x$")
        plt.title("Distribution of current density vs radius")
```

```
        plt.show()
        rad_fig.savefig(f"Derivative waveforms 2.5uA gain\\
            Derivative_of_2.5uA_gain{i}.png")

        signal_process(times, S, V, SNR, FWHM)

        plt.close(rad_fig)
```